

SEMESTER END EXAMINATIONS – AUGUST / SEPTEMBER 2023

Program	: B.E. - Information Science and Engineering	Semester	: IV
Course Name	: Design and Analysis of Algorithms	Max. Marks	: 100
Course Code	: IS44(O)	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

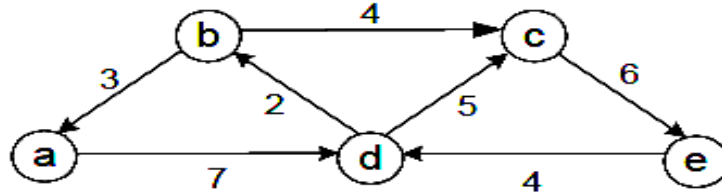
- Consider the following algorithm:
Algorithm MatMul(A[1...n],B[1..n],C[1...n])
//Multiplies two square matrices A and B
//Result: C[1...n].
for i=1 to n do
for j=1 to n do
C[i][j]=0
for k=1 to n do
C[i][j]=C[i][j]+A[i][k]*B[k][j]
return C
i)What is its basic operation?
ii)How many times is the basic operation executed?
iii)What is the efficiency class of this algorithm?
CO1 (08)
 - Explain in brief the basic Asymptotic notations and illustrate with figures.
CO1 (12)
- Discuss the various stages of algorithm design and analysis process using flow chart.
CO1 (10)
 - Write the general plan for analyzing time efficiency of recursive algorithms. Mathematically analyze factorial of a given number and find its time complexity.
CO1 (10)

UNIT - II

- Design a brute force algorithm for selection sort and Mathematically analyze its worst and best case efficiency classes.
CO2 (10)
 - With appropriate examples and shift tables, illustrate the four cases(situations)that occur while searching a pattern in the text using Horspool's Algorithm. Write code for Horspool's shift table as well as String Matching Algorithm.
CO2 (10)
- Design a recursive algorithm for Merge sort and analyze the efficiency of Merge sort for worst case time complexity using backward substitution method and also by applying Master's Theorem.
CO2 (10)
 - Discuss how Quick sort works to sort an array? Write an algorithm of Quick sort and trace for the following data set 'E','X','A','M','P','L','E'.
CO2 (10)

UNIT - III

5. a) Solve the following instance of Knapsack problem with the given capacity $W=8$ using dynamic programming. And also given the Number of items=4, Weight={2,3,4,5}, Profit={1,2,5,6} selected. CO3 (06)
- b) State and explain Dijkstra's algorithm to find single source shortest path for a given graph. Apply the same for the following graph. Assume "a" as the source vertex. CO3 (07)

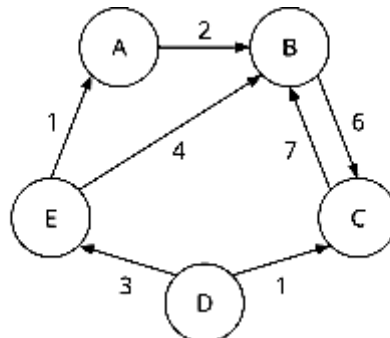


- c) i) Construct a Huffman code for the following data: CO3 (07)

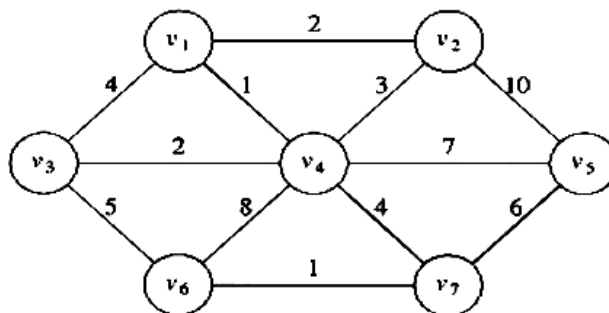
character	A	B	C	D	E
probability	0.4	0.1	0.2	0.14	0.16

- ii) Encode the text ABACABAD using the code.
 iii) Decode the text whose encoding is 100010111001010 using Huffman code.

6. a) Explain how dynamic programming is used to compute all pair's shortest paths for a weighted digraph, and apply the Floyd's algorithm to compute all pair shortest paths for the following graph. CO3 (10)



- b) Write the Kruskal's algorithm to find the minimum cost spanning tree. Trace the algorithm for the graph shown below. CO3 (10)



UNIT- IV

7. a) Define AVL tree. Explain the various types of rotations that can be performed to balance the tree. Construct AVL tree for the given list of elements: 60, 1, 10, 30, 100, 90, 80. CO4 (08)
- b) Explain the differences between DFS and BFS. CO4 (05)
- c) What is transform and conquer? Discuss the three variations of transform and conquer technique. CO4 (07)

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| 8. | a) | Design an algorithm to search an element using insertion sort and find its best, worst and average case time complexity. | CO4 | (08) |
| | b) | What is Josephus problem? Illustrate with an example the different methods of solving Josephus problem. Find J(9) using any one of the methods. | CO4 | (06) |
| | c) | Construct 2-3 tree for the list 9,5,8,3,2,4,7. | CO4 | (06) |

UNIT - V

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| 9. | a) | Differentiate between back tracking and branch-and-bound design techniques. | CO5 | (05) |
| | b) | How does backtracking design technique solve difficult combinatorial problems. Draw the complete state-space tree for solving 4-queens problem using backtracking. | CO5 | (10) |
| | c) | Explain heap and its vital properties. | CO5 | (05) |
| 10. | a) | Construct a heap for the list 1, 8, 6, 5, 3, 7, 4 by the bottom-up algorithm. Sort the contents of the list using heapsort. | CO5 | (10) |
| | b) | What is the need of approximation algorithms? How do you quantify the accuracy of an approximation algorithm? | CO5 | (05) |
| | c) | Define the following: (i) Tractable problems (ii) Class P (iii) Class NP (iv) Polynomial reduction (v) NP complete problems. | CO5 | (05) |
